

Empirical Proof Record: VALIDATED

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1. Claim

Across 30 synthetic (x, y) pairs of size 100 (seed=20260518), “scipy.stats.pearsonr“, “numpy.corrcoef“, and “pingouin.corr(method='pearson')“ produce Pearson r values that agree to $\leq 1e-09$ across all three pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer; three-way variant.)

- **Operationalisation:** Generate N (x, y) pairs where $x \sim \text{Normal}(0, 1)$ and $y = \rho_{\text{target}} \times x + \sqrt{1 - \rho_{\text{target}}^2} \times \text{Normal}(0, 1)$ for a sweep of ρ_{target} across $[-0.9, 0.9]$. Compute Pearson r under each of scipy, numpy, pingouin. Compute $\text{max_abs_diff} = \max \text{ over } (\text{pair, backend-comparison}) \text{ of } |r_a - r_b|$. VALIDATED iff $\text{max_abs_diff} \leq \text{tolerance}$ across all three pairwise comparisons.
- **Threshold:** `max_absolute_pearson_difference <= 1e-09 correlation`
- **H0:** At least one backend pair (scipy↔numpy, scipy↔pingouin, numpy↔pingouin) disagrees on Pearson r by more than the floating-point tolerance.
- **H1:** All three backends compute identical Pearson r to floating-point tolerance across every pair.

2. Verdict

VALIDATED — observed 3.3306690738754696e-16

30 pairs × 3 backends; max pairwise $|\Delta r| = 3.331e-16$ on scipy_vs_numpy ($\leq 1e-09$ tol)

3. Evidence

Pillar	Statistic	Value
pearson_scipy_vs_numpy_vs_pingouin	max_absolute_pearson_difference	3.3306690738754696e-1

4. Pre-registration

- Registered at: 2026-05-24T03:30:06.003730+00:00
- Config hash: 395ced3ae94d681473ffeddcaa298fd2...
- Data hash: 395ced3ae94d681473ffeddcaa298fd2...

5. Signature

87e55cfb2898dd4ace1f022e1476e4e027c93dcf45148706d413b62d8c577270